Enhanced Quadrature Encoder Pulse (eQEP) Module

The enhanced quadrature encoder pulse (eQEP) module is used for direct interface with a linear or rotary incremental encoder to get position, direction, and speed information from a rotating machine for use in a high-performance motion and position-control system. This microcontroller implements 2 instances of the eQEP module.

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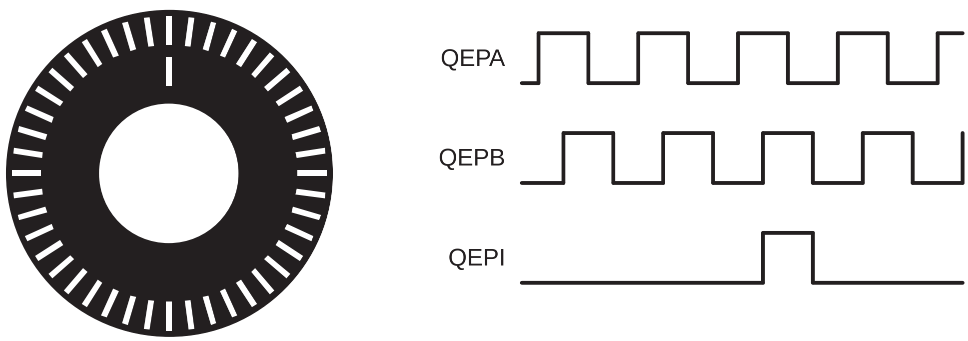
21.1 Introduction

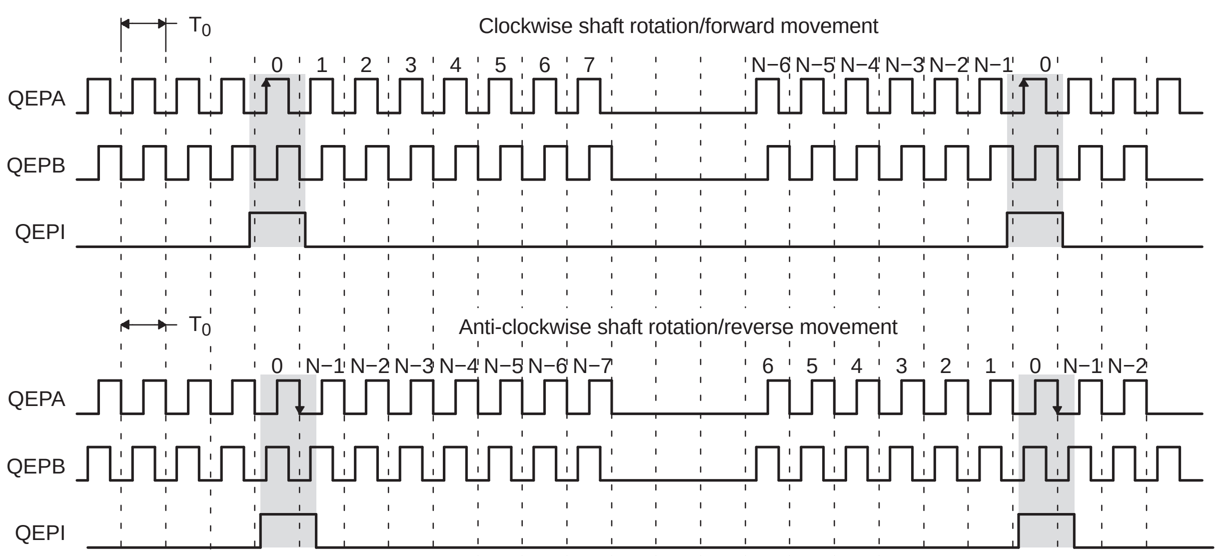
A single track of slots patterns the periphery of an incremental encoder disk, as shown in Figure 21-1. These slots create an alternating pattern of dark and light lines. The disk count is defined as the number of dark/light line pairs that occur per revolution (lines per revolution). As a rule, a second track is added to generate a signal that occurs once per revolution (index signal: QEPI), which can be used to indicate an absolute position. Encoder manufacturers identify the index pulse using different terms such as index, marker, home position, and zero reference

Figure 21-1. Optical Encoder Disk



To derive direction information, the lines on the disk are read out by two different photo-elements that "look" at the disk pattern with a mechanical shift of 1/4 the pitch of a line pair between them. This shift is realized with a reticle or mask that restricts the view of the photo-element to the desired part of the disk lines. As the disk rotates, the two photo-elements generate signals that are shifted 90° out of phase from each other. These are commonly called the quadrature QEPA and QEPB signals. The clockwise direction for most encoders is defined as the QEPA channel going positive before the QEPB channel and vise versa as shown in Figure 21-2.

Figure 21-2. QEP Encoder Output Signal for Forward/Reverse Movement

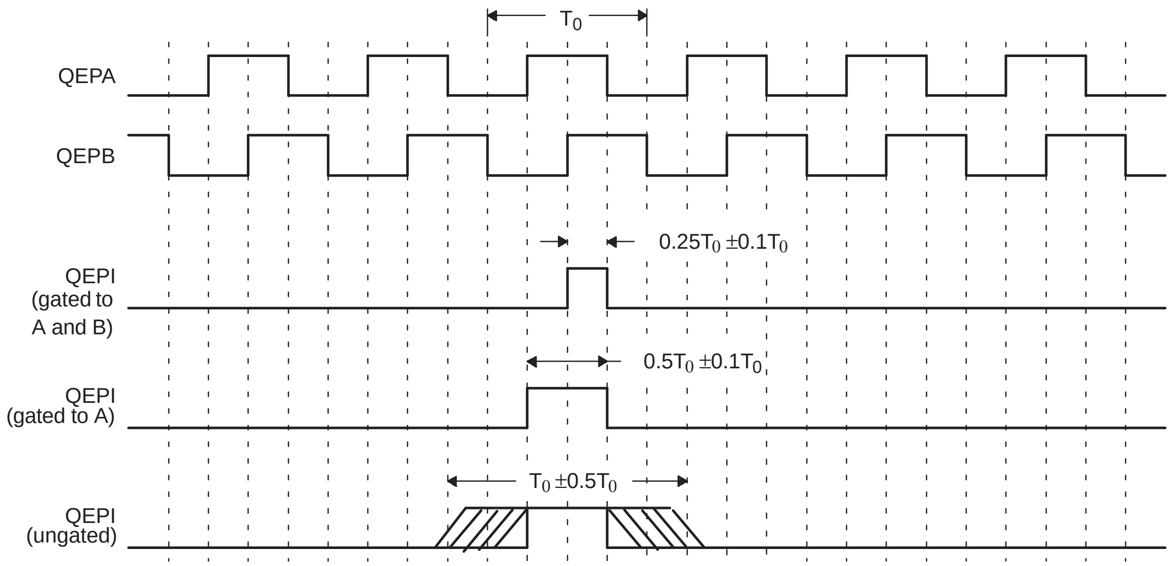


Legend: N = lines per revolution

The encoder wheel typically makes one revolution for every revolution of the motor or the wheel may be at a geared rotation ratio with respect to the motor. Therefore, the frequency of the digital signal coming from the QEPA and QEPB outputs varies proportionally with the velocity of the motor. For example, a 2000-line encoder directly coupled to a motor running at 5000 revolutions per minute (rpm) results in a frequency of 166.6 KHz, so by measuring the frequency of either the QEPA or QEPB output, the processor can determine the velocity of the motor.

Quadrature encoders from different manufacturers come with two forms of index pulse (gated index pulse or ungated index pulse) as shown in Figure 21-3. A nonstandard form of index pulse is ungated. In the ungated configuration, the index edges are not necessarily coincident with A and B signals. The gated index pulse is aligned to any of the four quadrature edges and width of the index pulse and can be equal to a quarter, half, or full period of the quadrature signal.

Figure 21-3. Index Pulse Example



Some typical applications of shaft encoders include robotics and even computer input in the form of a mouse. Inside your mouse you can see where the mouse ball spins a pair of axles (a left/right, and an up/down axle). These axles are connected to optical shaft encoders that effectively tell the computer how fast and in what direction the mouse is moving.

General Issues: Estimating velocity from a digital position sensor is a cost-effective strategy in motor control. Two different first order approximations for velocity may be written as:

$$v(k) \approx \frac{x(k) - x(k-1)}{T} = \frac{\Delta X}{T} \tag{27}$$

$$v(k) \approx \frac{X}{t(k) - t(k-1)} = \frac{X}{\Delta T} \tag{28}$$

where

v(k): Velocity at time instant k

x(k): Position at time instant k

x(k-1): Position at time instant k-1

T: Fixed unit time or inverse of velocity calculation rate

ΔX: Incremental position movement in unit time

t(k): Time instant "k"

t(k-1): Time instant "k-1"

X: Fixed unit position

ΔT: Incremental time elapsed for unit position movement.

Equation 27 is the conventional approach to velocity estimation and it requires a time base to provide unit time event for velocity calculation. Unit time is basically the inverse of the velocity calculation rate.

The encoder count (position) is read once during each unit time event. The quantity [x(k) - x(k-1)] is formed by subtracting the previous reading from the current reading. Then the velocity estimate is computed by multiplying by the known constant 1/T (where T is the constant time between unit time events and is known in advance).

Estimation based on Equation 27 has an inherent accuracy limit directly related to the resolution of the position sensor and the unit time period T. For example, consider a 500-line per revolution quadrature encoder with a velocity calculation rate of 400 Hz. When used for position the quadrature encoder gives a four-fold increase in resolution, in this case, 2000 counts per revolution. The minimum rotation that can be detected is therefore 0.0005 revolutions, which gives a velocity resolution of 12 rpm when sampled at 400 Hz. While this resolution may be satisfactory at moderate or high speeds, for example, 1% error at 1200 rpm, it would clearly prove inadequate at low speeds. In fact, at speeds below 12 rpm, the speed estimate would erroneously be zero much of the time.

At low speed, Equation 28 provides a more accurate approach. It requires a position sensor that outputs a fixed interval pulse train, such as the aforementioned quadrature encoder. The width of each pulse is defined by motor speed for a given sensor resolution. Equation 28 can be used to calculate motor speed by measuring the elapsed time between successive quadrature pulse edges. However, this method suffers from the opposite limitation, as does Equation 27. A combination of relatively large motor speeds and high sensor resolution makes the time interval ΔT small, and thus more greatly influenced by the timer resolution. This can introduce considerable error into high-speed estimates.

For systems with a large speed range (that is, speed estimation is needed at both low and high speeds), one approach is to use Equation 28 at low speed and have the DSP software switch over to Equation 27 when the motor speed rises above some specified threshold.

21.2 Description

This section provides the eQEP inputs, memory map, and functional description.

21.2.1 EQEP Inputs

The eQEP inputs include two pins for quadrature-clock mode or direction-count mode, an index (or 0 marker), and a strobe input.

- *QEPA/XCLK and QEPB/XDIR*

  These two pins can be used in quadrature-clock mode or direction-count mode.

  - *Quadrature-clock Mode*

    The eQEP encoders provide two square wave signals (A and B) 90 electrical degrees out of phase whose phase relationship is used to determine the direction of rotation of the input shaft and number of eQEP pulses from the index position to derive the relative position information. For forward or clockwise rotation, QEPA signal leads QEPB signal and vice versa. The quadrature decoder uses these two inputs to generate quadrature-clock and direction signals.

  - *Direction-count Mode*

    In direction-count mode, direction and clock signals are provided directly from the external source. Some position encoders have this type of output instead of quadrature output. The QEPA pin provides the clock input and the QEPB pin provides the direction input.

- *eQEPI: Index or Zero Marker*

  The eQEP encoder uses an index signal to assign an absolute start position from which position information is incrementally encoded using quadrature pulses. This pin is connected to the index output of the eQEP encoder to optionally reset the position counter for each revolution. This signal can be used to initialize or latch the position counter on the occurrence of a desired event on the index pin.

- QEPS: *Strobe Input*

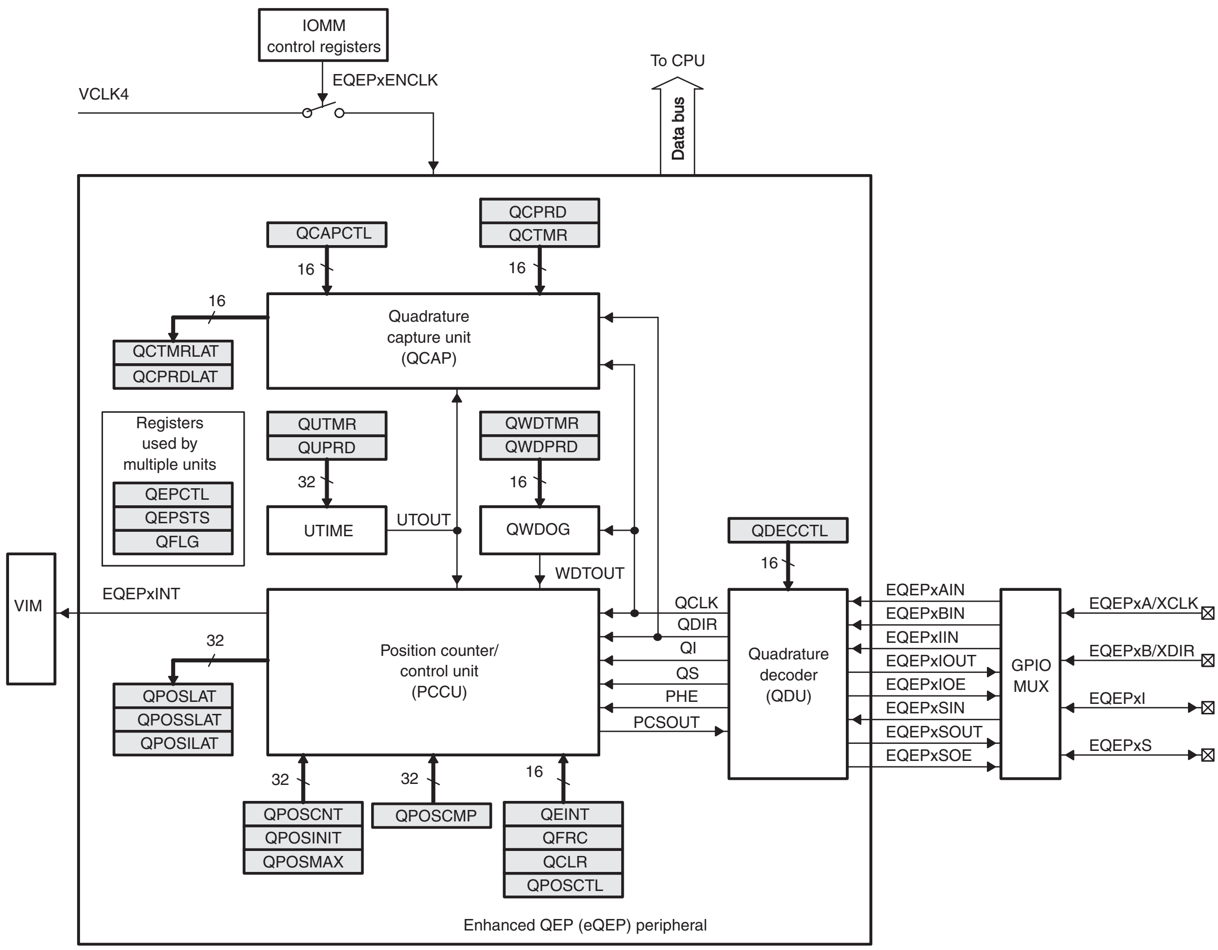
  This general-purpose strobe signal can initialize or latch the position counter on the occurrence of a desired event on the strobe pin. This signal is typically connected to a sensor or limit switch to notify that the motor has reached a defined position.

21.2.2 Functional Description

The eQEP peripheral contains the following major functional units (as shown in Figure 21-4):

- Programmable input qualification for each pin (part of the GPIO MUX)
- Quadrature decoder unit (QDU)
- Position counter and control unit for position measurement (PCCU)
- Quadrature edge-capture unit for low-speed measurement (QCAP)
- Unit time base for speed/frequency measurement (UTIME)
- Watchdog timer for detecting stalls (QWDOG)

Figure 21-4. Functional Block Diagram of the eQEP Peripheral



21.2.3 eQEP Memory Map

Table 21-1 lists the registers with their memory locations, sizes, and reset values.

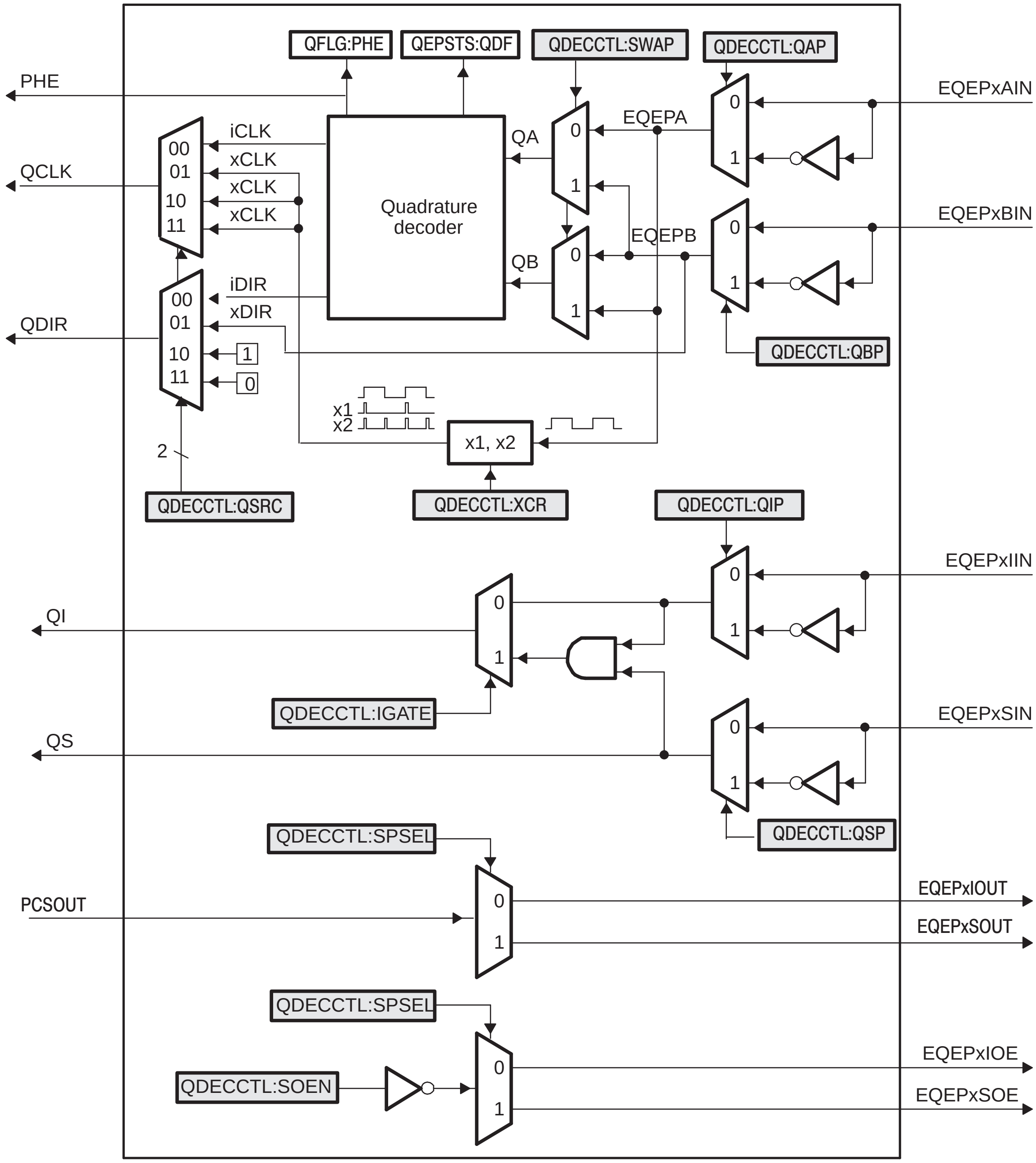
Table 21-1. EQEP Memory Map

Name	Address Offset	Size(x16)/ #shadow	Reset	Register Description
QPOSCNT	0x00	2/0	0x00000000	eQEP Position Counter
QPOSINIT	0x04	2/0	0x00000000	eQEP Initialization Position Count
QPOSMAX	0x08	2/0	0x00000000	eQEP Maximum Position Count
QPOSCMP	0x0C	2/1	0x00000000	eQEP Position-compare
QPOSILAT	0x10	2/0	0x00000000	eQEP Index Position Latch
QPOSSLAT	0x14	2/0	0x00000000	eQEP Strobe Position Latch
QPOSLAT	0x18	2/0	0x00000000	eQEP Position Latch
QUTMR	0x1C	2/0	0x00000000	eQEP Unit Timer
QUPRD	0x20	2/0	0x00000000	eQEP Unit Period Register
QWDTMR	0x24	1/0	0x0000	eQEP Watchdog Timer
QWDPRD	0x26	1/0	0x0000	eQEP Watchdog Period Register
QDECCTL	0x28	1/0	0x0000	eQEP Decoder Control Register
QEPCTL	0x2A	1/0	0x0000	eQEP Control Register
QCAPCTL	0x2C	1/0	0x0000	eQEP Capture Control Register
QPOSCTL	0x2E	1/0	0x00000	eQEP Position-compare Control Register
QEINT	0x30	1/0	0x0000	eQEP Interrupt Enable Register
QFLG	0x32	1/0	0x0000	eQEP Interrupt Flag Register
QCLR	0x34	1/0	0x0000	eQEP Interrupt Clear Register
QFRC	0x36	1/0	0x0000	eQEP Interrupt Force Register
QEPSTS	0x38	1/0	0x0000	eQEP Status Register
QCTMR	0x3A	1/0	0x0000	eQEP Capture Timer
QCPRD	0x3C	1/0	0x0000	eQEP Capture Period Register
QCTMRLAT	0x3E	1/0	0x0000	eQEP Capture Timer Latch
QCPRDLAT	0x40	1/0	0x0000	eQEP Capture Period Latch
Reserved	0x42	–	–	Reserved

21.3 Quadrature Decoder Unit (QDU)

Figure 21-5 shows a functional block diagram of the QDU.

Figure 21-5. Functional Block Diagram of Decoder Unit



21.3.1 Position Counter Input Modes

Clock and direction input to position counter is selected using QDECCTL[QSRC] bits, based on interface input requirement as follows:

- Quadrature-count mode
- Direction-count mode
- UP-count mode
- DOWN-count mode

21.3.1.1 Quadrature Count Mode

The quadrature decoder generates the direction and clock to the position counter in quadrature count mode.

Direction Decoding— The direction decoding logic of the eQEP circuit determines which one of the sequences (QEPA, QEPB) is the leading sequence and accordingly updates the direction information in QEPSTS[QDF] bit. Table 21-2 and Figure 21-6 show the direction decoding logic in truth table and state machine form. Both edges of the QEPA and QEPB signals are sensed to generate count pulses for the position counter. Therefore, the frequency of the clock generated by the eQEP logic is four times that of each input sequence. Figure 21-7 shows the direction decoding and clock generation from the eQEP input signals.

Phase Error Flag— In normal operating conditions, quadrature inputs QEPA and QEPB will be 90 degrees out of phase. The phase error flag (PHE) is set in the QFLG register when edge transition is detected simultaneously on the QEPA and QEPB signals to optionally generate interrupts. State transitions marked by dashed lines in Figure 21-6 are invalid transitions that generate a phase error.

Count Multiplication— The eQEP position counter provides 4x times the resolution of an input clock by generating a quadrature-clock (QCLK) on the rising/falling edges of both eQEP input clocks (QEPA and QEPB) as shown in Figure 21-7.

Reverse Count— In normal quadrature count operation, QEPA input is fed to the QA input of the quadrature decoder and the QEPB input is fed to the QB input of the quadrature decoder. Reverse counting is enabled by setting the SWAP bit in the QDECCTL register. This will swap the input to the quadrature decoder thereby reversing the counting direction.

Table 21-2. Quadrature Decoder Truth Table

Previous Edge	Present Edge	QDIR	QPOSCNT
QA↑	QB↑	UP	Increment
	QB↓	DOWN	Decrement
	QA↓	TOGGLE	Increment or Decrement
QA↓	QB↓	UP	Increment
	QB↑	DOWN	Decrement
	QA↑	TOGGLE	Increment or Decrement
QB↑	QA↑	DOWN	Increment
	QA↓	UP	Decrement
	QB↓	TOGGLE	Increment or Decrement
QB↓	QA↓	DOWN	Increment
	QA↑	UP	Decrement
	QB↑	TOGGLE	Increment or Decrement

Figure 21-6. Quadrature Decoder State Machine

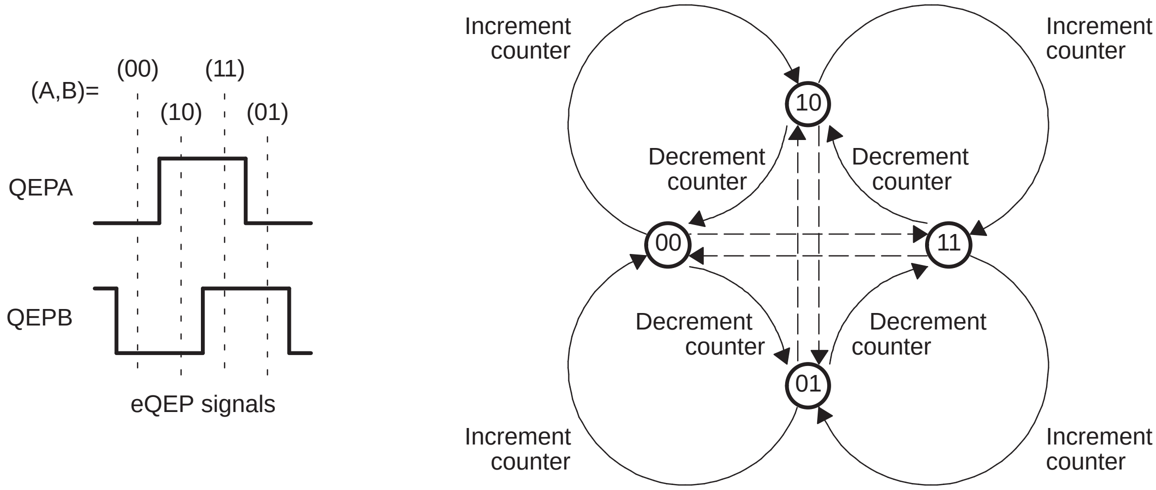
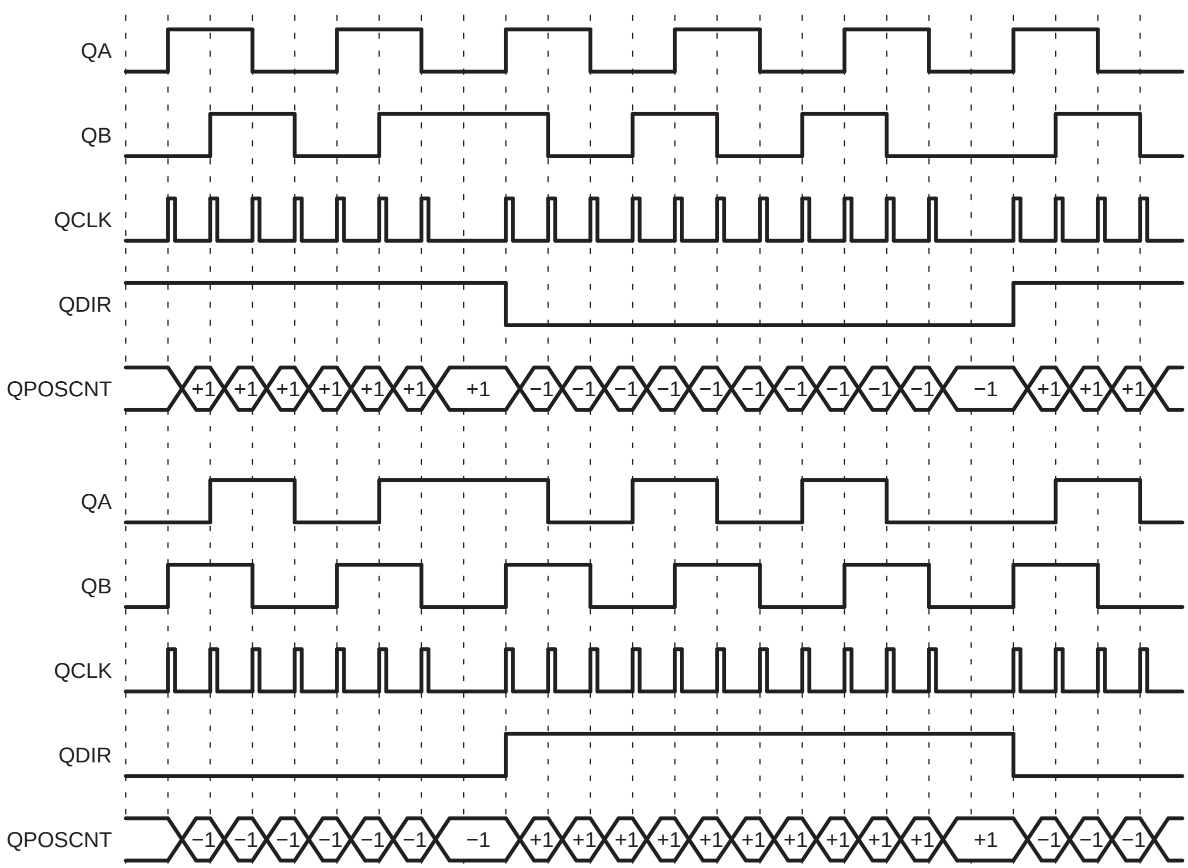


Figure 21-7. Quadrature-clock and Direction Decoding



21.3.1.2 Direction-Count Mode

Some position encoders provide direction and clock outputs, instead of quadrature outputs. In such cases, direction-count mode can be used. QEPA input will provide the clock for position counter and the QEPB input will have the direction information. The position counter is incremented on every rising edge of a QEPA input when the direction input is high and decremented when the direction input is low.

21.3.1.3 Up-Count Mode

The counter direction signal is hard-wired for up count and the position counter is used to measure the frequency of the QEPA input. Setting of the QDECCTL[XCR] bit enables clock generation to the position counter on both edges of the QEPA input, thereby increasing the measurement resolution by 2x factor.

21.3.1.4 Down-Count Mode

The counter direction signal is hardwired for a down count and the position counter is used to measure the frequency of the QEPA input. Setting of the QDECCTL[XCR] bit enables clock generation to the position counter on both edges of a QEPA input, thereby increasing the measurement resolution by 2x factor.

21.3.2 eQEP Input Polarity Selection

Each eQEP input can be inverted using QDECCTL[8:5] control bits. As an example, setting of QDECCTL[QIP] bit will invert the index input.

21.3.3 Position-Compare Sync Output

The enhanced eQEP peripheral includes a position-compare unit that is used to generate the position-compare sync signal on compare match between the position counter register (QPOSCNT) and the position-compare register (QPOSCMP). This sync signal can be output using an index pin or strobe pin of the EQEP peripheral.

Setting the QDECCTL[SOEN] bit enables the position-compare sync output and the QDECCTL[SPSEL] bit selects either an eQEP index pin or an eQEP strobe pin.

21.4 Position Counter and Control Unit (PCCU)

The position counter and control unit provides two configuration registers (QEPCTL and QPOSCTL) for setting up position counter operational modes, position counter initialization/latch modes and position-compare logic for sync signal generation.

21.4.1 Position Counter Operating Modes

Position counter data may be captured in different manners. In some systems, the position counter is accumulated continuously for multiple revolutions and the position counter value provides the position information with respect to the known reference. An example of this is the quadrature encoder mounted on the motor controlling the print head in the printer. Here the position counter is reset by moving the print head to the home position and then position counter provides absolute position information with respect to home position.

In other systems, the position counter is reset on every revolution using index pulse and position counter provides rotor angle with respect to index pulse position.

Position counter can be configured to operate in following four modes

- Position Counter Reset on Index Event
- Position Counter Reset on Maximum Position
- Position Counter Reset on the first Index Event
- Position Counter Reset on Unit Time Out Event (Frequency Measurement)

In all the above operating modes, position counter is reset to 0 on overflow and to QPOSMAX register value on underflow. Overflow occurs when the position counter counts up after QPOSMAX value. Underflow occurs when position counter counts down after "0". Interrupt flag is set to indicate overflow/underflow in QFLG register.

21.4.1.1 Position Counter Reset on Index Event (QEPCTL[PCRM] = 00)

If the index event occurs during the forward movement, then position counter is reset to 0 on the next eQEP clock. If the index event occurs during the reverse movement, then the position counter is reset to the value in the QPOSMAX register on the next eQEP clock.

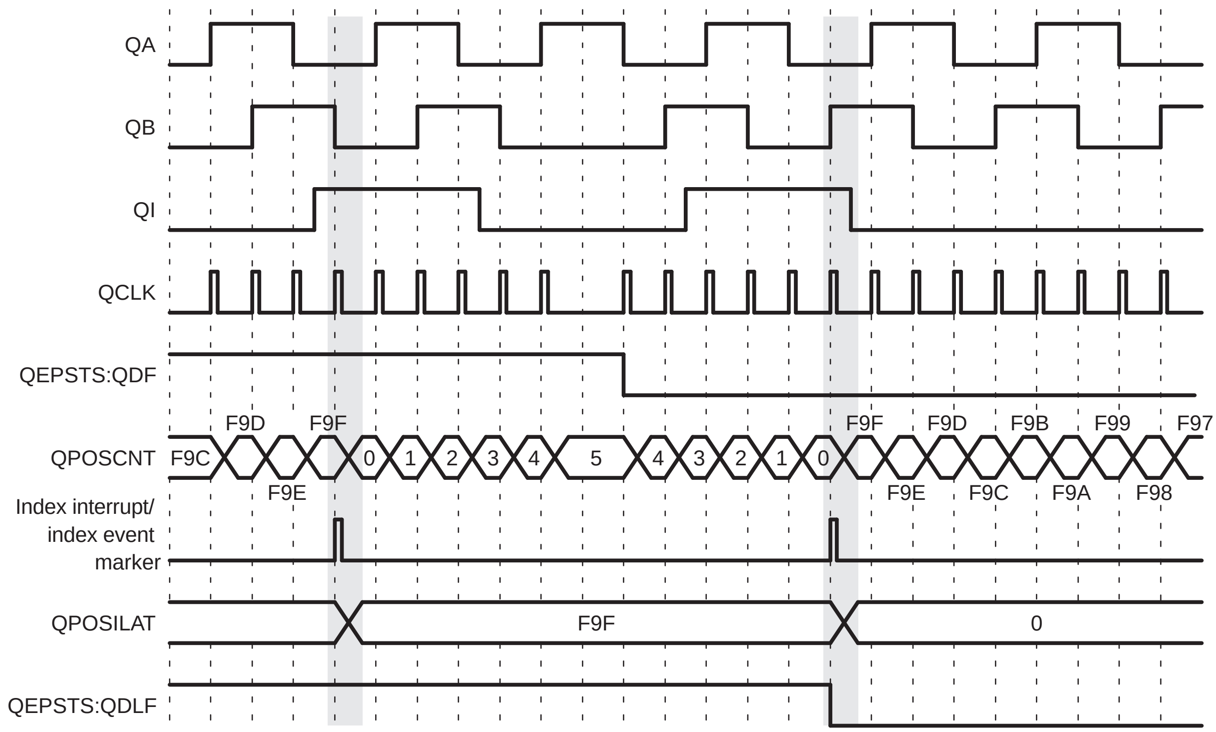
First index marker is defined as the quadrature edge following the first index edge. The eQEP peripheral records the occurrence of the first index marker (QEPSTS[FIMF]) and direction on the first index event marker (QEPSTS[FIDF]) in QEPSTS registers, it also remembers the quadrature edge on the first index marker so that same relative quadrature transition is used for index event reset operation.

For example, if the first reset operation occurs on the falling edge of QEPB during the forward direction, then all the subsequent reset must be aligned with the falling edge of QEPB for the forward rotation and on the rising edge of QEPB for the reverse rotation as shown in Figure 21-8.

The position-counter value is latched to the QPOSILAT register and direction information is recorded in the QEPSTS[QDLF] bit on every index event marker. The position-counter error flag (QEPSTS[PCEF]) and error interrupt flag (QFLG[PCE]) are set if the latched value is not equal to 0 or QPOSMAX. The position-counter error flag (QEPSTS[PCEF]) is updated on every index event marker and an interrupt flag (QFLG[PCE]) will be set on error that can be cleared only through software.

The index event latch configuration QEPCTL[IEL] bits are ignored in this mode and position counter error flag/interrupt flag are generated only in index event reset mode.

Figure 21-8. Position Counter Reset by Index Pulse for 1000 Line Encoder (QPOSMAX = 3999 or 0xF9F)

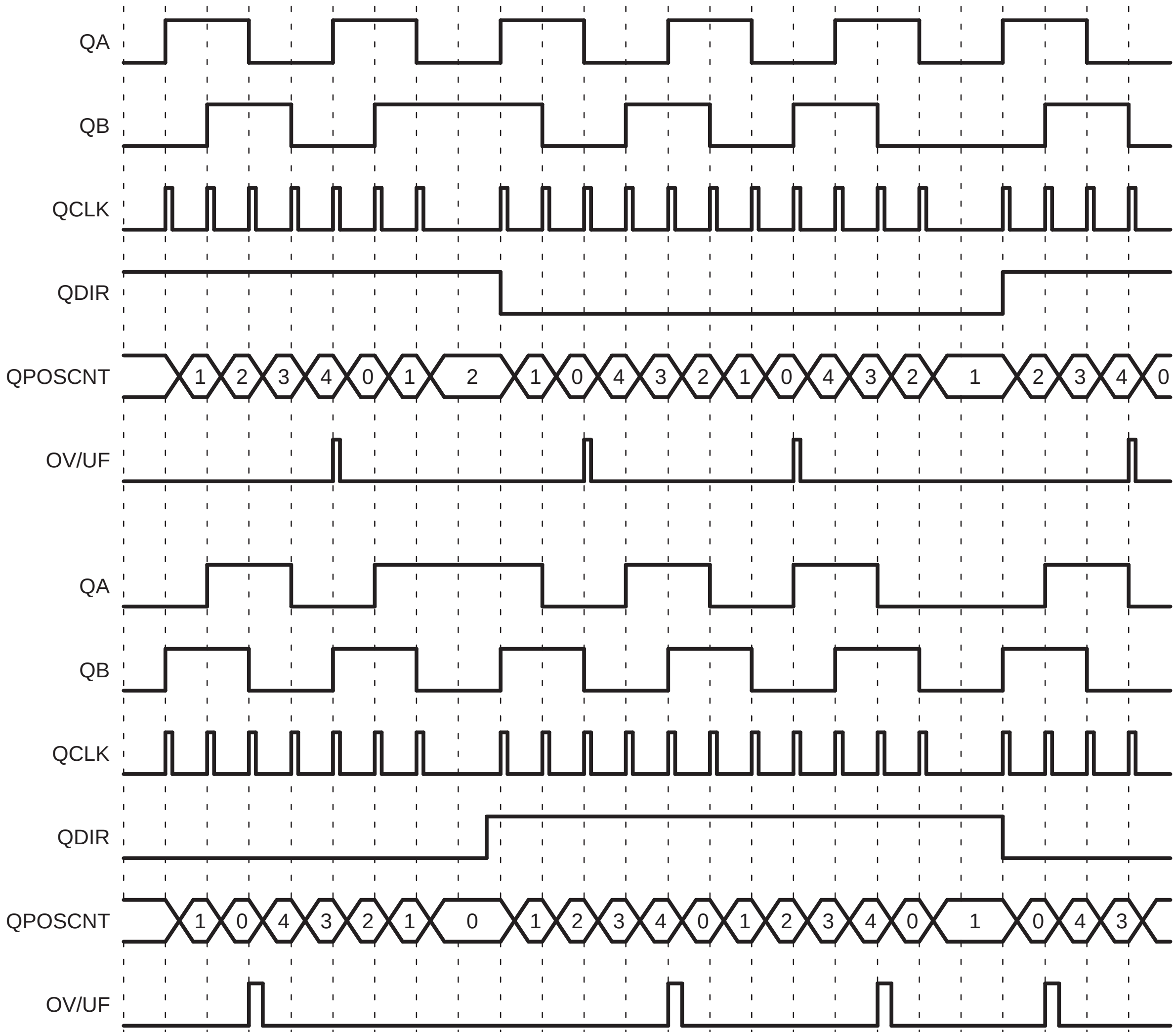


21.4.1.2 Position Counter Reset on Maximum Position (QEPCTL[PCRM] = 01)

If the position counter is equal to QPOSMAX, then the position counter is reset to 0 on the next eQEP clock for forward movement and position counter overflow flag is set. If the position counter is equal to ZERO, then the position counter is reset to QPOSMAX on the next QEP clock for reverse movement and position counter underflow flag is set. Figure 21-9 shows the position counter reset operation in this mode.

First index marker is defined as the quadrature edge following the first index edge. The eQEP peripheral records the occurrence of the first index marker (QEPSTS[FIMF]) and direction on the first index event marker (QEPSTS[FIDF]) in the QEPSTS registers; it also remembers the quadrature edge on the first index marker so that the same relative quadrature transition is used for the software index marker (QEPCTL[IEL]=11).

Figure 21-9. Position Counter Underflow/Overflow (QPOSMAX = 4)



21.4.1.3 Position Counter Reset on the First Index Event (QEPCTL[PCRM] = 10)

If the index event occurs during forward movement, then the position counter is reset to 0 on the next eQEP clock. If the index event occurs during the reverse movement, then the position counter is reset to the value in the QPOSMAX register on the next eQEP clock. Note that this is done only on the first occurrence and subsequently the position counter value is not reset on an index event; rather, it is reset based on maximum position as described in Section 21.4.1.2.

First index marker is defined as the quadrature edge following the first index edge. The eQEP peripheral records the occurrence of the first index marker (QEPSTS[FIMF]) and direction on the first index event marker (QEPSTS[FIDF]) in QEPSTS registers, it also remembers the quadrature edge on the first index marker so that same relative quadrature transition is used for software index marker (QEPCTL[IEL]=11).

21.4.1.4 Position Counter Reset on Unit Time out Event (QEPCTL[PCRM] = 11)

In this mode, the QPOSCNT value is latched to the QPOSLAT register and then the QPOSCNT is reset (to 0 or QPOSMAX, depending on the direction mode selected by QDECCTL[QSRC] bits on a unit time event). This is useful for frequency measurement.

21.4.2 Position Counter Latch

The eQEP index and strobe input can be configured to latch the position counter (QPOSCNT) into QPOSILAT and QPOSSLAT, respectively, on occurrence of a definite event on these pins.

21.4.2.1 Index Event Latch

In some applications, it may not be desirable to reset the position counter on every index event and instead it may be required to operate the position counter in full 32-bit mode (QEPCTL[PCRM] = 01 and QEPCTL[PCRM] = 10 modes).

In such cases, the eQEP position counter can be configured to latch on the following events and direction information is recorded in the QEPSTS[QDLF] bit on every index event marker.

- Latch on Rising edge (QEPCTL[IEL]=01)
- Latch on Falling edge (QEPCTL[IEL]=10)
- Latch on Index Event Marker (QEPCTL[IEL]=11)

This is particularly useful as an error checking mechanism to check if the position counter accumulated the correct number of counts between index events. As an example, the 1000-line encoder must count 4000 times when moving in the same direction between the index events.

The index event latch interrupt flag (QFLG[IEL]) is set when the position counter is latched to the QPOSILAT register. The index event latch configuration bits (QEPCTZ[IEL]) are ignored when QEPCTL[PCRM] = 00.

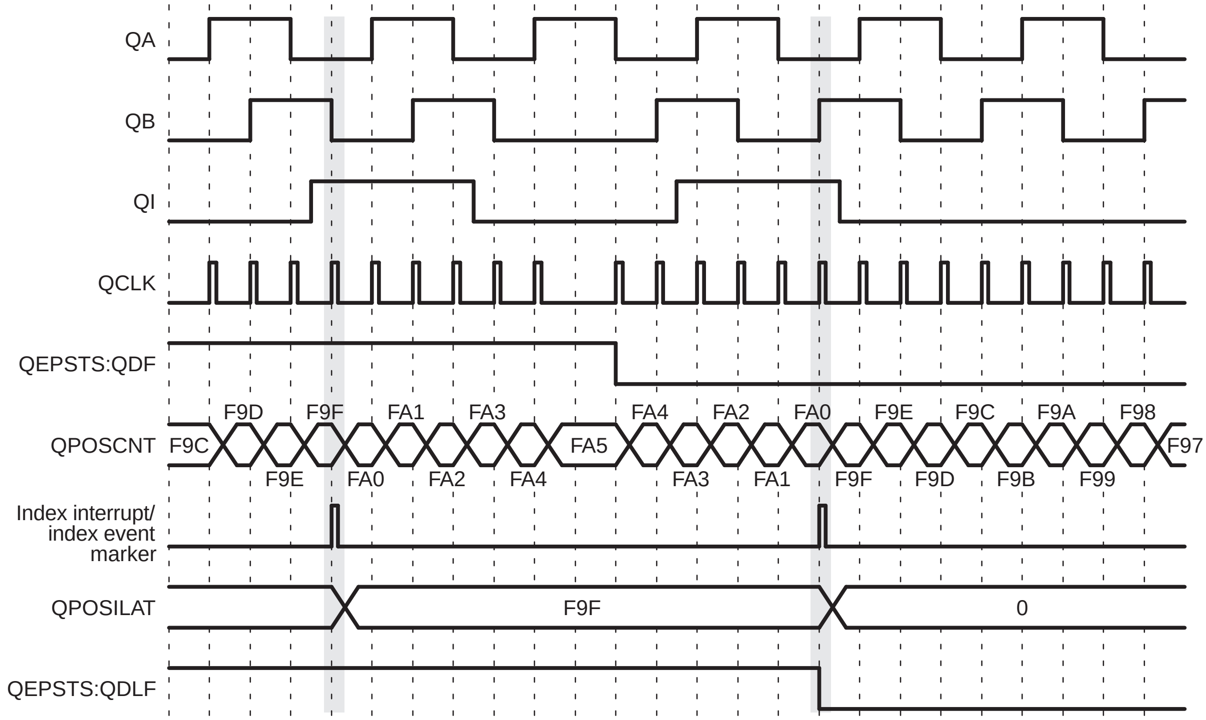
Latch on Rising Edge (QEPCTL[IEL]=01)— The position counter value (QPOSCNT) is latched to the QPOSILAT register on every rising edge of an index input.

Latch on Falling Edge (QEPCTL[IEL] = 10)— The position counter value (QPOSCNT) is latched to the QPOSILAT register on every falling edge of index input.

Latch on Index Event Marker/Software Index Marker (QEPCTL[IEL] = 11— The first index marker is defined as the quadrature edge following the first index edge. The eQEP peripheral records the occurrence of the first index marker (QEPSTS[FIMF]) and direction on the first index event marker (QEPSTS[FIDF]) in the QEPSTS registers. It also remembers the quadrature edge on the first index marker so that same relative quadrature transition is used for latching the position counter (QEPCTL[IEL]=11).

Figure 21-10 shows the position counter latch using an index event marker.

Figure 21-10. Software Index Marker for 1000-line Encoder (QEPCTL[IEL] = 1)



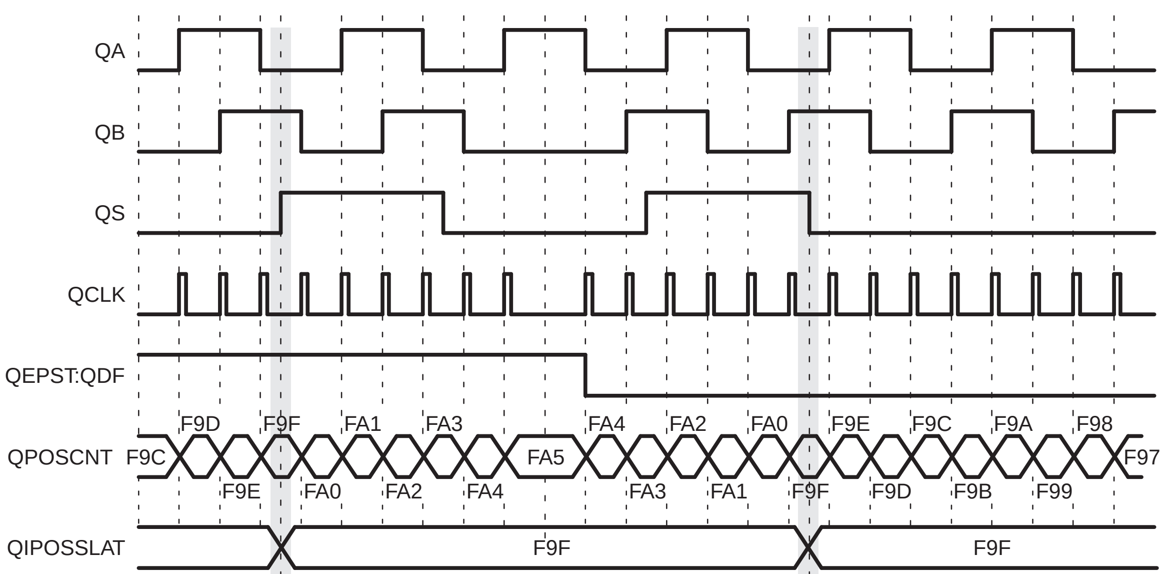
21.4.2.2 Strobe Event Latch

The position-counter value is latched to the QPOSSLAT register on the rising edge of the strobe input by clearing the QEPCTL[SEL] bit.

If the QEPCTL[SEL] bit is set, then the position counter value is latched to the QPOSSLAT register on the rising edge of the strobe input for forward direction and on the falling edge of the strobe input for reverse direction as shown in Figure 21-11.

The strobe event latch interrupt flag (QFLG[SEL) is set when the position counter is latched to the QPOSSLAT register.

Figure 21-11. Strobe Event Latch (QEPCTL[SEL] = 1)



21.4.3 Position Counter Initialization

The position counter can be initialized using following events:

- Index event
- Strobe event
- Software initialization

Index Event Initialization (IEI)— The QEPI index input can be used to trigger the initialization of the position counter at the rising or falling edge of the index input. If the QEPCTL[IEI] bits are 10, then the position counter (QPOSCNT) is initialized with a value in the QPOSINIT register on the rising edge of index input. Conversely, if the QEPCTL[IEI] bits are 11, initialization will be on the falling edge of the index input.

Strobe Event Initialization (SEI)— If the QEPCTL[SEI] bits are 10, then the position counter is initialized with a value in the QPOSINIT register on the rising edge of strobe input.

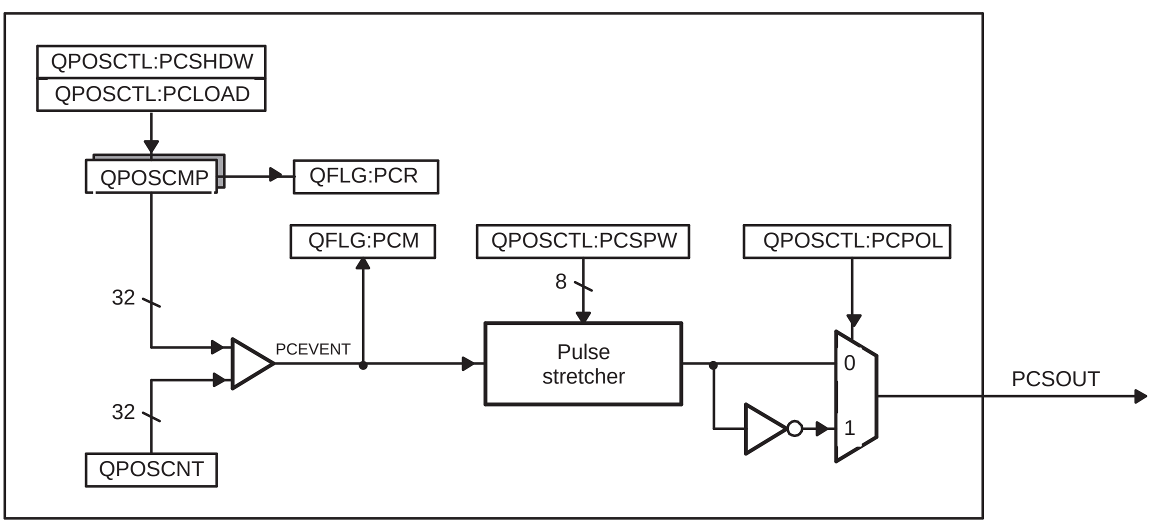
If QEPCTL[SEL] bits are 11, then the position counter is initialized with a value in the QPOSINIT register on the rising edge of strobe input for forward direction and on the falling edge of strobe input for reverse direction.

Software Initialization (SWI)— The position counter can be initialized in software by writing a 1 to the QEPCTL[SWI] bit. This bit is not automatically cleared. While the bit is still set, if a 1 is written to it again, the position counter will be re-initialized.

21.4.4 eQEP Position-compare Unit

The eQEP peripheral includes a position-compare unit that is used to generate a sync output and/or interrupt on a position-compare match. Figure 21-12 shows a diagram. The position-compare (QPOSCMP) register is shadowed and shadow mode can be enabled or disabled using the QPOSCTL[PSSHDW] bit. If the shadow mode is not enabled, the CPU writes directly to the active position compare register.

Figure 21-12. eQEP Position-compare Unit



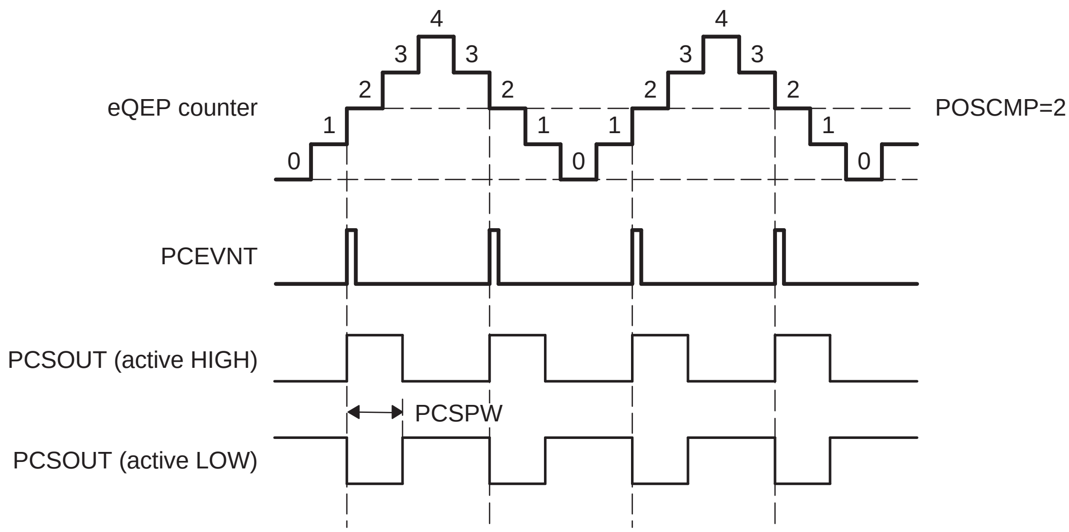
In shadow mode, you can configure the position-compare unit (QPOSCTL[PCLOAD]) to load the shadow register value into the active register on the following events and to generate the position-compare ready (QFLG[PCR]) interrupt after loading.

- Load on compare match
- Load on position-counter zero event

The position-compare match (QFLG[PCM]) is set when the position-counter value (QPOSCNT) matches with the active position-compare register (QPOSCMP) and the position-compare sync output of the programmable pulse width is generated on compare match to trigger an external device.

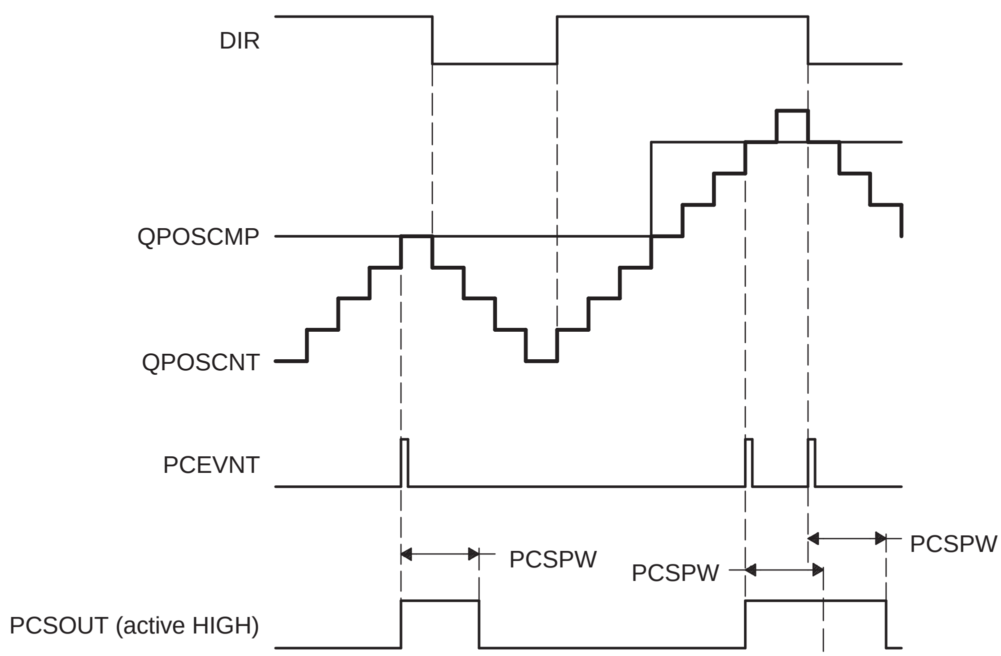
For example, if QPOSCMP = 2, the position-compare unit generates a position-compare event on 1 to 2 transitions of the eQEP position counter for forward counting direction and on 3 to 2 transitions of the eQEP position counter for reverse counting direction (see Figure 21-13).

Figure 21-13. eQEP Position-compare Event Generation Points



The pulse stretcher logic in the position-compare unit generates a programmable position-compare sync pulse output on the position-compare match. In the event of a new position-compare match while a previous position-compare pulse is still active, then the pulse stretcher generates a pulse of specified duration from the new position-compare event as shown in Figure 21-14.

Figure 21-14. eQEP Position-compare Sync Output Pulse Stretcher



21.5 eQEP Edge Capture Unit

The eQEP peripheral includes an integrated edge capture unit to measure the elapsed time between the unit position events as shown in Figure 21-15. This feature is typically used for low speed measurement using the following equation:

$$v(k) = \frac{X}{t(k) - t(k-1)} = \frac{X}{\Delta T} \tag{29}$$

where,

- X - Unit position is defined by integer multiple of quadrature edges (see Figure 21-16)
- ΔT - Elapsed time between unit position events
- v(k) - Velocity at time instant "k"

The eQEP capture timer (QCTMR) runs from prescaled VCLK4 and the prescaler is programmed by the QCAPCTL[CCPS] bits. The capture timer (QCTMR) value is latched into the capture period register (QCPRD) on every unit position event and then the capture timer is reset, a flag is set in QEPSTS:UPEVNT to indicate that new value is latched into the QCPRD register. Software can check this status flag before reading the period register for low speed measurement and clear the flag by writing 1.

Time measurement (ΔT) between unit position events will be correct if the following conditions are met:

- No more than 65,535 counts have occurred between unit position events.
- No direction change between unit position events.

The capture unit sets the eQEP overflow error flag (QEPSTS[COEF]) in the event of capture timer overflow between unit position events. If a direction change occurs between the unit position events, then an error flag is set in the status register (QEPSTS[CDEF]).

Capture Timer (QCTMR) and Capture period register (QCPRD) can be configured to latch on following events.

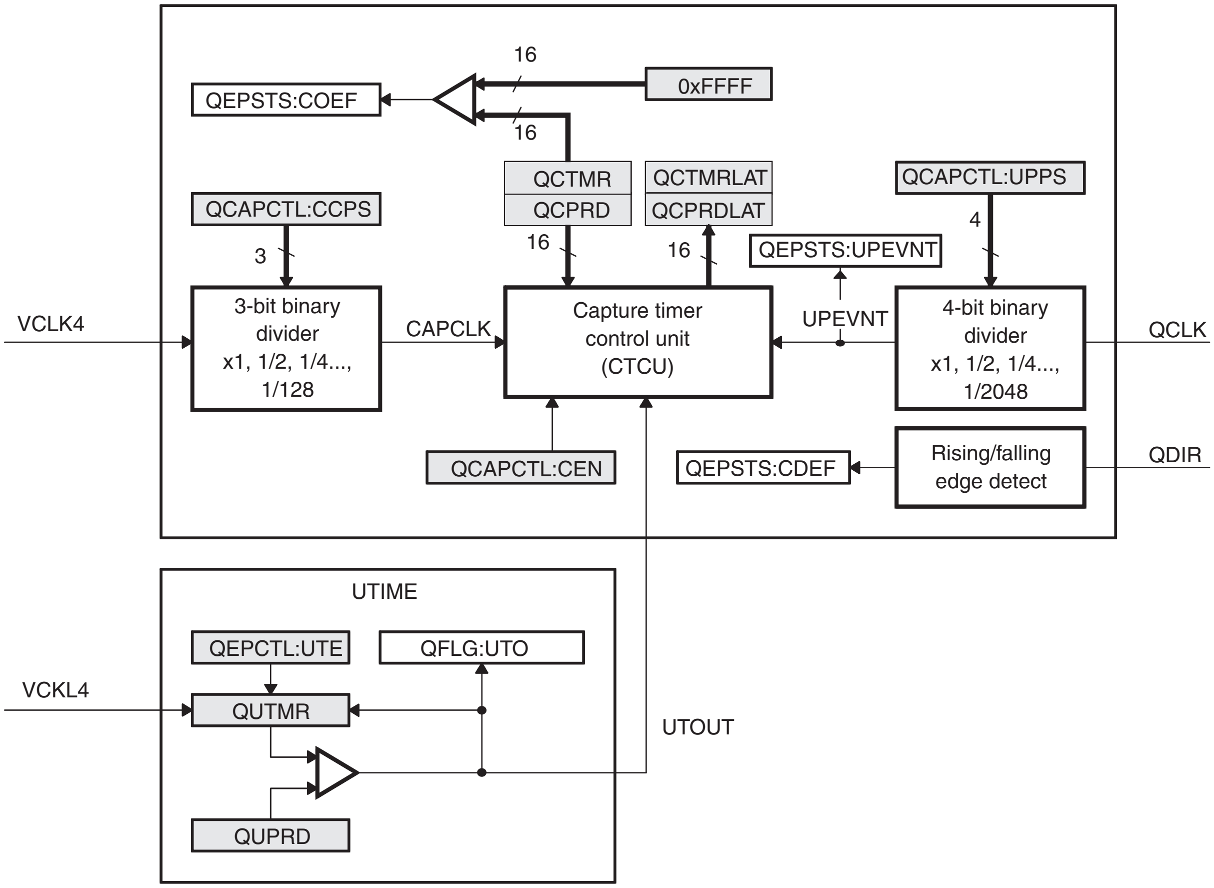
- CPU read of QPOSCNT register
- Unit time-out event

If the QEPCTL[QCLM] bit is cleared, then the capture timer and capture period values are latched into the QCTMRLAT and QCPRDLAT registers, respectively, when the CPU reads the position counter (QPOSCNT).

If the QEPCTL[QCLM] bit is set, then the position counter, capture timer, and capture period values are latched into the QPOSLAT, QCTMRLAT and QCPRDLAT registers, respectively, on unit time out.

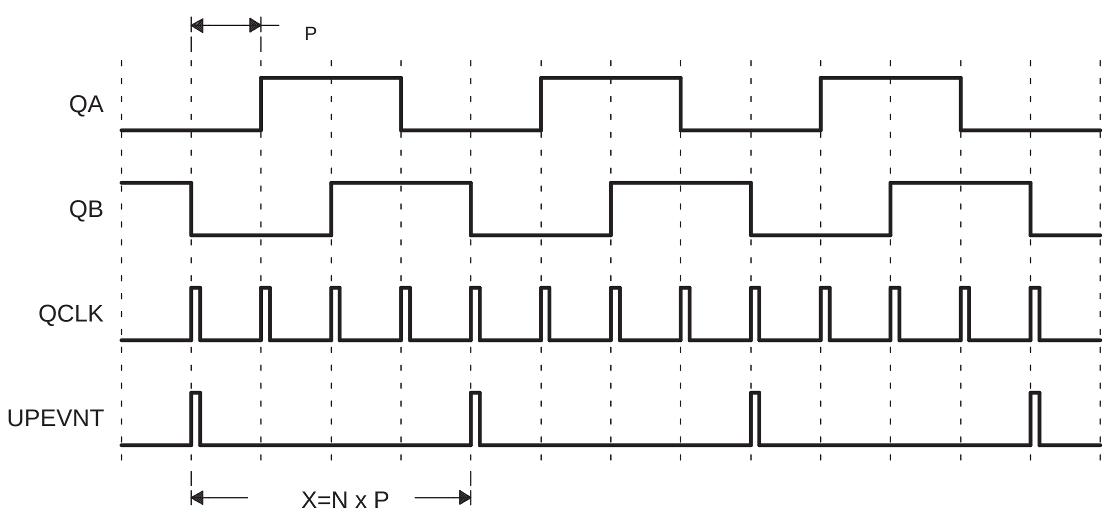
Figure 21-17 shows the capture unit operation along with the position counter.

Figure 21-15. eQEP Edge Capture Unit



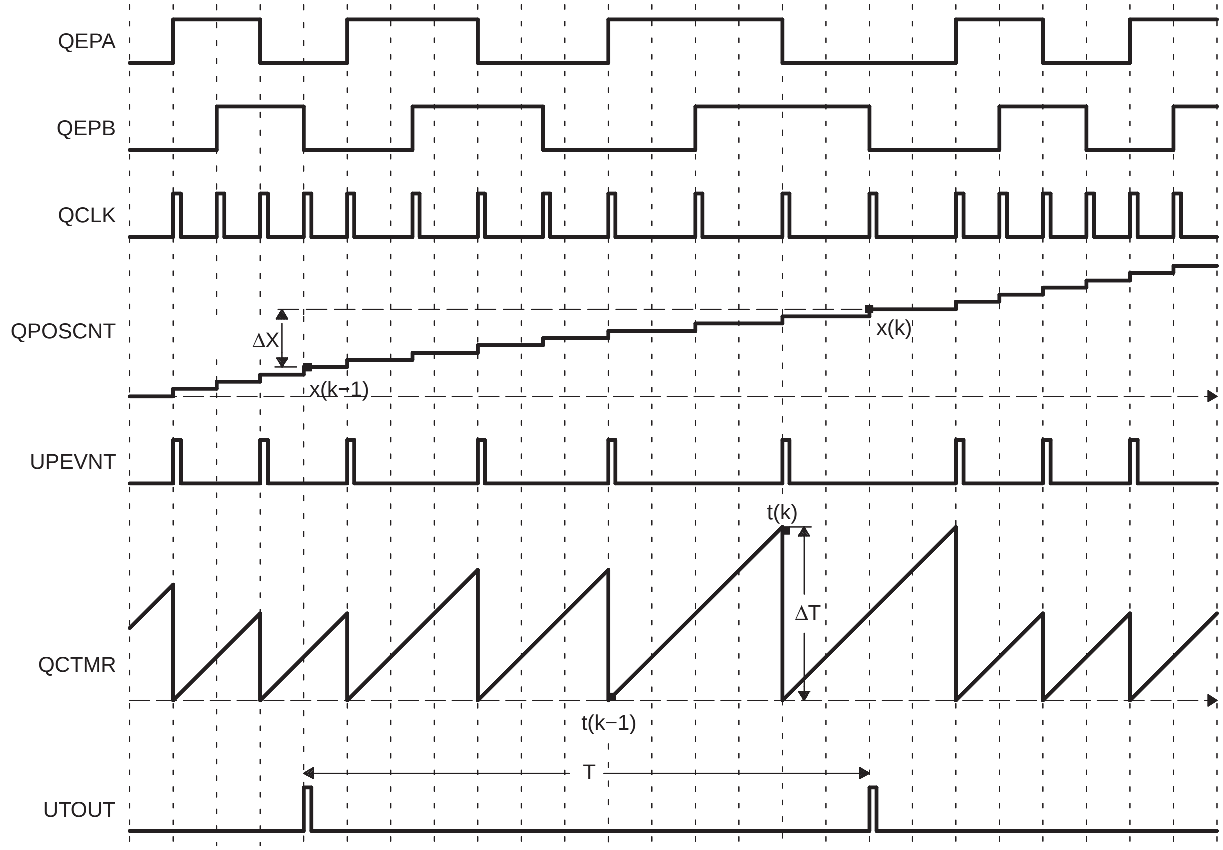
NOTE: The QCAPCTL[UPPS] prescaler should not be modified dynamically (such as switching the unit event prescaler from QCLK/4 to QCLK/8). Doing so may result in undefined behavior. The QCAPCTL[CPPS] prescaler can be modified dynamically (such as switching CAPCLK prescaling mode from SYSCLK/4 to SYSCLK/8) only after the capture unit is disabled.

Figure 21-16. Unit Position Event for Low Speed Measurement (QCAPCTL[UPPS] = 0010)



N = Number of quadrature periods selected using QCAPCTL[UPPS] bits

Figure 21-17. eQEP Edge Capture Unit - Timing Details



Velocity Calculation Equations:

$$v(k) = \frac{x(k) - x(k-1)}{T} = \frac{\Delta X}{T} \quad \text{o} \qquad (30)$$

where

v(k): Velocity at time instant k

x(k): Position at time instant k

x(k-1): Position at time instant k-1

T: Fixed unit time or inverse of velocity calculation rate

ΔX: Incremental position movement in unit time

X: Fixed unit position

ΔT: Incremental time elapsed for unit position movement

t(k): Time instant "k"

t(k-1): Time instant "k-1"

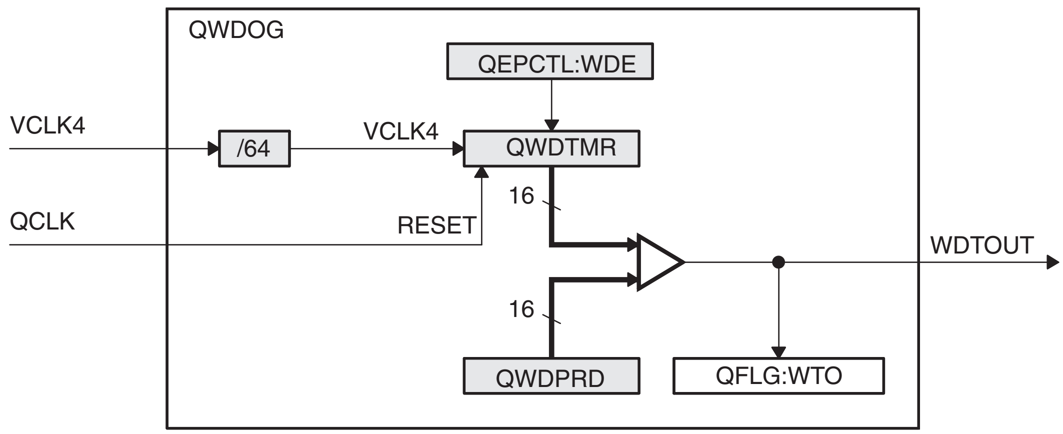
Unit time (T) and unit period(X) are configured using the QUPRD and QCAPCTL[UPPS] registers. Incremental position output and incremental time output is available in the QPOSLAT and QCPRDLAT registers.

Parameter	Relevant Register to Configure or Read the Information
T	Unit Period Register (QUPRD)
ΔX	Incremental Position = QPOSLAT(k) - QPOSLAT(K-1)
X	Fixed unit position defined by sensor resolution and ZCAPCTL[UPPS] bits
ΔT	Capture Period Latch (QCPRDLAT)

21.6 eQEP Watchdog

The eQEP peripheral contains a 16-bit watchdog timer that monitors the quadrature-clock to indicate proper operation of the motion-control system. The eQEP watchdog timer is clocked from VCLK4/64 and the quadrate clock event (pulse) resets the watchdog timer. If no quadrature-clock event is detected until a period match (QWDPRD = QWDTMR), then the watchdog timer will time out and the watchdog interrupt flag will be set (QFLG[WTO]). The time-out value is programmable through the watchdog period register (QWDPRD).

Figure 21-18. eQEP Watchdog Timer

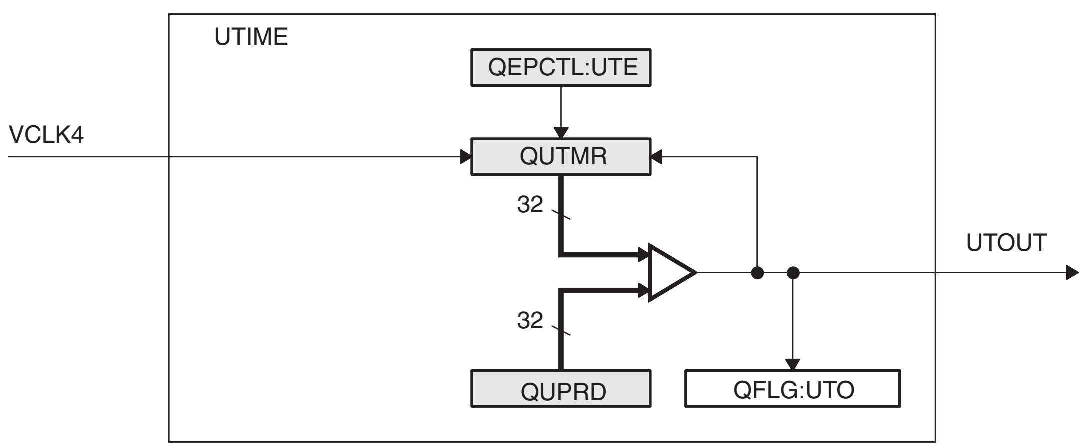


21.7 Unit Timer Base

The eQEP peripheral includes a 32-bit timer (QUTMR) that is clocked by VCLK4 to generate periodic interrupts for velocity calculations. The unit time out interrupt is set (QFLG[UTO]) when the unit timer (QUTMR) matches the unit period register (QUPRD).

The eQEP peripheral can be configured to latch the position counter, capture timer, and capture period values on a unit time out event so that latched values are used for velocity calculation as described in Section 21.5.

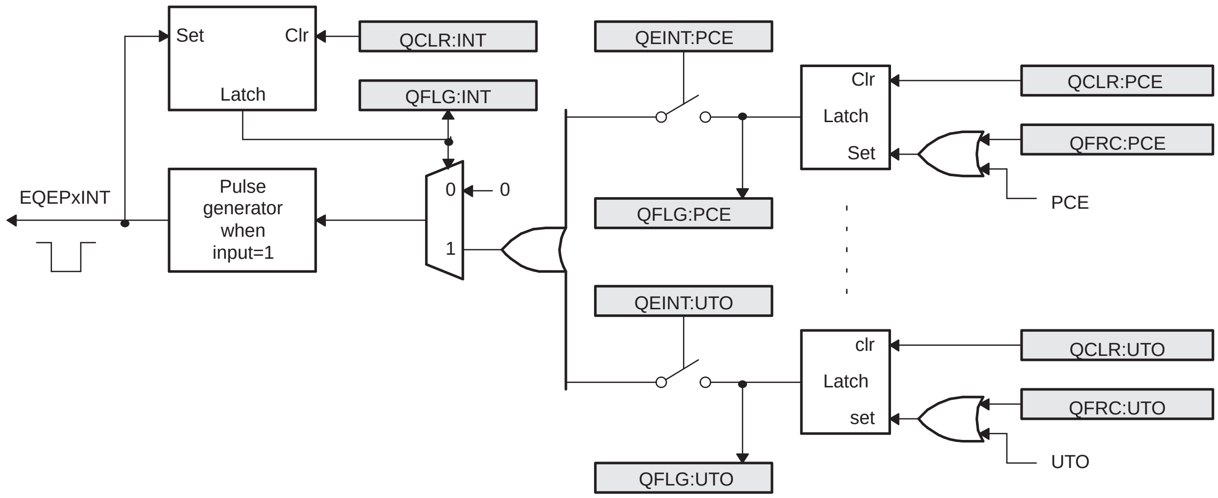
Figure 21-19. eQEP Unit Time Base



21.8 eQEP Interrupt Structure

Figure 21-20 shows how the interrupt mechanism works in the EQEP module.

Figure 21-20. EQEP Interrupt Generation



Eleven interrupt events (PCE, PHE, QDC, WTO, PCU, PCO, PCR, PCM, SEL, IEL and UTO) can be generated. The interrupt control register (QEINT) is used to enable/disable individual interrupt event sources. The interrupt flag register (QFLG) indicates if any interrupt event has been latched and contains the global interrupt flag bit (INT). An interrupt pulse is generated only to the PIE if any of the interrupt events is enabled, the flag bit is 1 and the INT flag bit is 0. The interrupt service routine will need to clear the global interrupt flag bit and the serviced event, via the interrupt clear register (QCLR), before any other interrupt pulses are generated. You can force an interrupt event by way of the interrupt force register (QFRC), which is useful for test purposes.

21.9 EQEP Registers

Table 21-3 lists the memory-mapped registers for the EQEP. All register offset addresses not listed in Table 21-3 should be considered as reserved locations and the register contents should not be modified.

The base address for the control registers is FCF7 9900h for eQEP1 and FCF7 9A00h for eQEP2.

Table 21-3. EQEP Registers

Offset	Acronym	Register Name	Section
0h	QPOSCNT	eQEP Position Counter Register	Section 21.9.1
4h	QPOSINIT	eQEP Position Counter Initialization Register	Section 21.9.2
8h	QPOSMAX	eQEP Maximum Position Count Register	Section 21.9.3
Ch	QPOSCMP	eQEP Position-compare Register	Section 21.9.4
10h	QPOSILAT	eQEP Index Position Latch Register	Section 21.9.5
14h	QPOSSLAT	eQEP Strobe Position Latch Register	Section 21.9.6
18h	QPOSLAT	eQEP Position Counter Latch Register	Section 21.9.7
1Ch	QUTMR	eQEP Unit Timer Register	Section 21.9.8
20h	QUPRD	eQEP Register Unit Period Register	Section 21.9.9
24h	QWDTMR	eQEP Watchdog Timer Register	Section 21.9.10
26h	QWDPRD	eQEP Watchdog Period Register	Section 21.9.11
28h	QDECCTL	eQEP Decoder Control Register	Section 21.9.12
2Ah	QEPCTL	eQEP Control Register	Section 21.9.13
2Ch	QCAPCTL	eQEP Capture Control Register	Section 21.9.14
2Eh	QPOSCTL	eQEP Position-compare Control Register	Section 21.9.15
30h	QEINT	eQEP Interrupt Enable Register	Section 21.9.16
32h	QFLG	eQEP Interrupt Flag Register	Section 21.9.17
34h	QCLR	eQEP Interrupt Clear Register	Section 21.9.18
36h	QFRC	eQEP Interrupt Force Register	Section 21.9.19
38h	QEPSTS	eQEP Status Register	Section 21.9.20
3Ah	QCTMR	eQEP Capture Timer Register	Section 21.9.21
3Ch	QCPRD	eQEP Capture Period Register	Section 21.9.22
3Eh	QCTMRLAT	eQEP Capture Timer Latch Register	Section 21.9.23
40h	QCPRDLAT	eQEP Capture Period Latch Register	Section 21.9.24

Complex bit access types are encoded to fit into small table cells. Table 21-4 shows the codes that are used for access types in this section.

Table 21-4. EQEP Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
Reset or Default Value		
-n		Value after reset or the default value

21.9.1 QPOSCNT Register (Offset = 0h) [reset = 0h]

QPOSCNT is shown in Figure 21-21 and described in Table 21-5.

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Figure 21-21. QPOSCNT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSCNT																															
R/W-0h																															

Table 21-5. QPOSCNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSCNT	R/W	0h	This 32-bit position counter register counts up/down on every eQEP pulse based on direction input. This counter acts as a position integrator whose count value is proportional to position from a give reference point.

21.9.2 QPOSINIT Register (Offset = 4h) [reset = 0h]

QPOSINIT is shown in Figure 21-22 and described in Table 21-6.

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Figure 21-22. QPOSINIT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSINIT																															
R/W-0h																															

Table 21-6. QPOSINIT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSINIT	R/W	0h	This register contains the position value that is used to initialize the position counter based on external strobe or index event. The position counter is initialized through software.

21.9.3 QPOSMAX Register (Offset = 8h) [reset = 0h]

QPOSMAX is shown in Figure 21-23 and described in Table 21-7.

Return to Summary Table.

Figure 21-23. QPOSMAX Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSMAX																															
R/W-0h																															

Table 21-7. QPOSMAX Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSMAX	R/W	0h	This register contains the maximum position counter value.

21.9.4 QPOSCMP Register (Offset = Ch) [reset = 0h]

QPOSCMP is shown in Figure 21-24 and described in Table 21-8.

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Figure 21-24. QPOSCMP Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSCMP																															
R/W-0h																															

Table 21-8. QPOSCMP Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSCMP	R/W	0h	The position-compare value in this register is compared with the position counter (QPOSCNT) to generate sync output and/or interrupt on compare match.

21.9.5 QPOSILAT Register (Offset = 10h) [reset = 0h]

QPOSILAT is shown in Figure 21-25 and described in Table 21-9.

Return to Summary Table.

Figure 21-25. QPOSILAT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSILAT																															
R-0h																															

Table 21-9. QPOSILAT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSILAT	R	0h	The position-counter value is latched into this register on an index event as defined by the QEPCTL[IEL] bits.

21.9.6 QPOSSLAT Register (Offset = 14h) [reset = 0h]

QPOSSLAT is shown in Figure 21-26 and described in Table 21-10.

Return to Summary Table.

Figure 21-26. QPOSSLAT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSSLAT																															
R-0h																															

Table 21-10. QPOSSLAT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSSLAT	R	0h	The position-counter value is latched into this register on strobe event as defined by the QEPCTL[SEL] bits.

21.9.7 QPOSLAT Register (Offset = 18h) [reset = 0h]

QPOSLAT is shown in Figure 21-27 and described in Table 21-11.

Return to Summary Table.

Figure 21-27. QPOSLAT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QPOSLAT																															
R-0h																															

Table 21-11. QPOSLAT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QPOSLAT	R	0h	The position-counter value is latched into this register on unit time out event.

21.9.8 QUTMR Register (Offset = 1Ch) [reset = 0h]

QUTMR is shown in Figure 21-28 and described in Table 21-12.

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Figure 21-28. QUTMR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QUTMR																															
R/W-0h																															

Table 21-12. QUTMR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QUTMR	R/W	0h	This register acts as time base for unit time event generation. When this timer value matches with unit time period value, unit time event is generated.

21.9.9 QUPRD Register (Offset = 20h) [reset = 0h]

QUPRD is shown in Figure 21-29 and described in Table 21-13.

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Figure 21-29. QUPRD Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QUPRD																															
R/W-0h																															

Table 21-13. QUPRD Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	QUPRD	R/W	0h	This register contains the period count for unit timer to generate periodic unit time events to latch the eQEP position information at periodic interval and optionally to generate interrupt.

21.9.10 QWDTMR Register (Offset = 24h) [reset = 0h]

QWDTMR is shown in Figure 21-30 and described in Table 21-14.

Return to Summary Table.

Figure 21-30. QWDTMR Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QWDTMR															
R/W-0h															

Table 21-14. QWDTMR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QWDTMR	R/W	0h	This register acts as time base for watchdog to detect motor stalls. When this timer value matches with watchdog period value, watchdog timeout interrupt is generated. This register is reset upon edge transition in quadrature-clock indicating the motion.

21.9.11 QWDPRD Register (Offset = 26h) [reset = 0h]

QWDPRD is shown in Figure 21-31 and described in Table 21-15.

Return to Summary Table.

Figure 21-31. QWDPRD Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QWDPRD															
R/W-0h															

Table 21-15. QWDPRD Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QWDPRD	R/W	0h	This register contains the time-out count for the eQEP peripheral watchdog timer. When the watchdog timer value matches the watchdog period value, a watchdog timeout interrupt is generated.

21.9.12 QDECCTL Register (Offset = 28h) [reset = 0h]

QDECCTL is shown in Figure 21-32 and described in Table 21-16.

Return to Summary Table.

Figure 21-32. QDECCTL Register

15	14	13	12	11	10	9	8
QSRC		SOEN	SPSEL	XCR	SWAP	IGATE	QAP
R/W-0h		R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
QBP	QIP	QSP	RESERVED				
R/W-0h	R/W-0h	R/W-0h	R-0h				

Table 21-16. QDECCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	QSRC	R/W	0h	Position-counter source selection. 0h = Quadrature count mode (QCLK = iCLK, QDIR = iDIR) 1h = Direction-count mode (QCLK = xCLK, QDIR = xDIR) 2h = UP count mode for frequency measurement (QCLK = xCLK, QDIR = 1) 3h = DOWN count mode for frequency measurement (QCLK = xCLK, QDIR = 0)
13	SOEN	R/W	0h	Sync output-enable. 0h = Disable position-compare sync output. 1h = Enable position-compare sync output.
12	SPSEL	R/W	0h	Sync output pin selection. 0h = Index pin is used for sync output. 1h = Strobe pin is used for sync output.
11	XCR	R/W	0h	External clock rate. 0h = 2x resolution: Count the rising/falling edge. 1h = 1x resolution: Count the rising edge only.
10	SWAP	R/W	0h	Swap quadrature clock inputs. This swaps the input to the quadrature decoder, reversing the counting direction. 0h = Quadrature-clock inputs are not swapped. 1h = Quadrature-clock inputs are swapped.
9	IGATE	R/W	0h	Index pulse gating option. 0h = Disable gating of Index pulse. 1h = Gate the index pin with strobe.
8	QAP	R/W	0h	QEPA input polarity. 0h = No effect. 1h = Negates QEPA input.
7	QBP	R/W	0h	QEPB input polarity. 0h = No effect. 1h = Negates QEPB input.
6	QIP	R/W	0h	QEPI input polarity. 0h = No effect. 1h = Negates QEPI input.
5	QSP	R/W	0h	QEPS input polarity. 0h = No effect. 1h = Negates QEPS input.
4-0	RESERVED	R	0h	Always write as 0.

21.9.13 QEPCTL Register (Offset = 2Ah) [reset = 0h]

QEPCTL is shown in Figure 21-33 and described in Table 21-17.

Return to Summary Table.

Figure 21-33. QEPCTL Register

15	14	13	12	11	10	9	8
FREE_SOFT		PCRM		SEI		IEI	
R/W-0h		R/W-0h		R/W-0h		R/W-0h	
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
SWI	SEL	IEL		QPEN	QCLM	UTE	WDE
R/W-0h	R/W-0h	R/W-0h		R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 21-17. QEPCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-14	FREE_SOFT	R/W	0h	Emulation control bits. QPOSCNT behavior: 0x0 = Position counter stops immediately on emulation suspend. 0x1 = Position counter continues to count until the rollover. 0x2 = Position counter is unaffected by emulation suspend. 0x3 = Position counter is unaffected by emulation suspend. QWDTMR behavior: 0x0 = Watchdog counter stops immediately. 0x1 = Watchdog counter counts until WD period match roll over. 0x2 = Watchdog counter is unaffected by emulation suspend. 0x3 = Watchdog counter is unaffected by emulation suspend. QUTMR behavior: 0x0 = Unit timer stops immediately. 0x1 = Unit timer counts until period rollover. 0x2 = Unit timer is unaffected by emulation suspend. 0x3 = Unit timer is unaffected by emulation suspend. QCTMR behavior: 0x0 = Capture Timer stops immediately. 0x1 = Capture Timer counts until next unit period event. 0x2 = Capture Timer is unaffected by emulation suspend. 0x3 = Capture Timer is unaffected by emulation suspend.
13-12	PCRM	R/W	0h	Position counter reset mode. 0h = Position counter reset on an index event. 1h = Position counter reset on the maximum position. 2h = Position counter reset on the first index event. 3h = Position counter reset on a unit time event.
11-10	SEI	R/W	0h	Strobe event initialization of position counter. 0h = Does nothing (action is disabled). 1h = Does nothing (action is disabled). 2h = Initializes the position counter on rising edge of the QEPS signal . 3h = Clockwise Direction: Initializes the position counter on the rising edge of QEPS strobe. Counter Clockwise Direction: Initializes the position counter on the falling edge of QEPS strobe.
9-8	IEI	R/W	0h	Index event initialization of position counter. 0h = Do nothing (action is disabled). 1h = Do nothing (action is disabled). 2h = Initializes the position counter on the rising edge of the QEPI signal (QPOSCNT = QPOSINIT). 3h = Initializes the position counter on the falling edge of QEPI signal (QPOSCNT = QPOSINIT).

Table 21-17. QEPCTL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7	SWI	R/W	0h	Software initialization of position counter. 0h = Do nothing (action is disabled). 1h = Initialize position counter (QPOSCNT = QPOSINIT). This bit is not cleared automatically.
6	SEL	R/W	0h	Strobe event latch of position counter. 0h = The position counter is latched on the rising edge of QEPS strobe (QPOSSLAT = POSCCNT). Latching on the falling edge is done by inverting the strobe input using the QSP bit in the QDECCTL register. 1h = Clockwise Direction: Position counter is latched on rising edge of QEPS strobe. Counter Clockwise Direction: Position counter is latched on falling edge of QEPS strobe.
5-4	IEL	R/W	0h	Index event latch of position counter (software index marker). 0h = Reserved 1h = Latches position counter on rising edge of the index signal. 2h = Latches position counter on falling edge of the index signal. 3h = Software index marker. Latches the position counter and quadrature direction flag on index event marker. The position counter is latched to the QPOSILAT register and the direction flag is latched in the QEPSTS[QDLF] bit. This mode is useful for software index marking.
3	QPEN	R/W	0h	Quadrature position counter enable/software reset. 0h = Reset the eQEP peripheral internal operating flags/read-only registers. Control/configuration registers are not disturbed by a software reset. 1h = eQEP position counter is enabled.
2	QCLM	R/W	0h	eQEP capture latch mode. 0h = Latch on position counter read by CPU. Capture timer and capture period values are latched into QCTMRLAT and QCPRDLAT registers when CPU reads the QPOSCNT register. 1h = Latch on unit time out. Position counter, capture timer and capture period values are latched into QPOSLAT, QCTMRLAT, and QCPRDLAT registers on unit time out.
1	UTE	R/W	0h	eQEP unit timer enable. 0h = Disable eQEP unit timer. 1h = Enable eQEP unit timer.
0	WDE	R/W	0h	eQEP watchdog enable. 0h = Disable eQEP watchdog timer. 1h = Enable eQEP watchdog timer.

21.9.14 QCAPCTL Register (Offset = 2Ch) [reset = 0h]

QCAPCTL is shown in Figure 21-34 and described in Table 21-18.

Return to Summary Table.

Figure 21-34. QCAPCTL Register

15	14	13	12	11	10	9	8
CEN	RESERVED						
R/W-0h	R-0h						
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
RESERVED	CCPS			UPPS			
R-0h	R/W-0h			R/W-0h			

Table 21-18. QCAPCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	CEN	R/W	0h	eQEP capture enable. 0h = eQEP capture unit is disabled. 1h = eQEP capture unit is enabled.
14-7	RESERVED	R	0h	Always write as 0.
6-4	CCPS	R/W	0h	eQEP capture timer clock prescaler. 0h = CAPCLK = VCLK4/1 1h = CAPCLK = VCLK4/2 2h = CAPCLK = VCLK4/4 3h = CAPCLK = VCLK4/8 4h = CAPCLK = VCLK4/16 5h = CAPCLK = VCLK4/32 6h = CAPCLK = VCLK4/64 7h = CAPCLK = VCLK4/128
3-0	UPPS	R/W	0h	Unit position event prescaler. 0h = UPEVNT = QCLK/1 1h = UPEVNT = QCLK/2 2h = UPEVNT = QCLK/4 3h = UPEVNT = QCLK/8 4h = UPEVNT = QCLK/16 5h = UPEVNT = QCLK/32 6h = UPEVNT = QCLK/64 7h = UPEVNT = QCLK/128 8h = UPEVNT = QCLK/256 9h = UPEVNT = QCLK/512 Ah = UPEVNT = QCLK/1024 Bh = UPEVNT = QCLK/2048 Ch = Reserved Dh = Reserved Eh = Reserved Fh = Reserved

21.9.15 QPOSCTL Register (Offset = 2Eh) [reset = 0h]

QPOSCTL is shown in Figure 21-35 and described in Table 21-19.

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Figure 21-35. QPOSCTL Register

15	14	13	12	11	10	9	8
PCSHDW	PCLOAD	PCPOL	PCE	PCSPW			
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h			
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PCSPW							
R/W-0h							

Table 21-19. QPOSCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	PCSHDW	R/W	0h	Position-compare shadow enable. 0h = Shadow is disabled, load Immediate. 1h = Shadow is enabled.
14	PCLOAD	R/W	0h	Position-compare shadow load mode. 0h = Load on QPOSCNT = 0. 1h = Load when QPOSCNT = QPOSCMP.
13	PCPOL	R/W	0h	Polarity of sync output. 0h = Active HIGH pulse output. 1h = Active LOW pulse output.
12	PCE	R/W	0h	Position-compare enable. 0h = Disable position compare unit. 1h = Enable position compare unit.
11-0	PCSPW	R/W	0h	Select-position-compare sync output pulse width. Valid values: 0 to FFFh. 0h = 1 x 4 x VCLK4 cycles 1h = 2 x 4 x VCLK4 cycles FFFh = 4096 x 4 x VCLK4 cycles

21.9.16 QEINT Register (Offset = 30h) [reset = 0h]

QEINT is shown in Figure 21-36 and described in Table 21-20.

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Figure 21-36. QEINT Register

15	14	13	12	11	10	9	8
RESERVED				UTO	IEL	SEL	PCM
R-0h				R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PCR	PCO	PCU	WTO	QDC	QPE	PCE	RESERVED
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 21-20. QEINT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Always write as 0
11	UTO	R/W	0h	Unit time out interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
10	IEL	R/W	0h	Index event latch interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
9	SEL	R/W	0h	Strobe event latch interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
8	PCM	R/W	0h	Position-compare match interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
7	PCR	R/W	0h	Position-compare ready interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
6	PCO	R/W	0h	Position counter overflow interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
5	PCU	R/W	0h	Position counter underflow interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
4	WTO	R/W	0h	Watchdog time out interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
3	QDC	R/W	0h	Quadrature direction change interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
2	QPE	R/W	0h	Quadrature phase error interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
1	PCE	R/W	0h	Position counter error interrupt enable. 0h = Interrupt is disabled. 1h = Interrupt is enabled.
0	RESERVED	R	0h	Reserved

21.9.17 QFLG Register (Offset = 32h) [reset = 0h]

QFLG is shown in Figure 21-37 and described in Table 21-21.

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Figure 21-37. QFLG Register

15	14	13	12	11	10	9	8
RESERVED				UTO	IEL	SEL	PCM
R-0h				R-0h	R-0h	R-0h	R-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PCR	PCO	PCU	WTO	QDC	PHE	PCE	INT
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 21-21. QFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Always write as 0.
11	UTO	R	0h	Unit time out interrupt flag. 0h = No interrupt is generated. 1h = Set by eQEP unit timer period match.
10	IEL	R	0h	Index event latch interrupt flag. 0h = No interrupt is generated. 1h = Set after latching the QPOSCNT to QPOSILAT.
9	SEL	R	0h	Strobe event latch interrupt flag. 0h = No interrupt is generated. 1h = Set after latching the QPOSCNT to QPOSSLAT.
8	PCM	R	0h	eQEP compare match event interrupt flag. 0h = No interrupt is generated. 1h = Set on position-compare match.
7	PCR	R	0h	Position-compare ready interrupt flag. 0h = No interrupt is generated. 1h = Set after transferring the shadow register value to the active position compare register.
6	PCO	R	0h	Position counter overflow interrupt flag. 0h = No interrupt is generated. 1h = Set on position counter overflow.
5	PCU	R	0h	Position counter underflow interrupt flag. 0h = No interrupt is generated. 1h = Set on position counter underflow.
4	WTO	R	0h	Watchdog timeout interrupt flag. 0h = No interrupt is generated. 1h = Set by watchdog timeout.
3	QDC	R	0h	Quadrature direction change interrupt flag. 0h = No interrupt is generated. 1h = Set during change of direction.
2	PHE	R	0h	Quadrature phase error interrupt flag. 0h = No interrupt is generated. 1h = Set on simultaneous transition of QEPA and QEPB.
1	PCE	R	0h	Position counter error interrupt flag. 0h = No interrupt is generated. 1h = Position counter error.

Table 21-21. QFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	INT	R	0h	Global interrupt status flag. 0h = No interrupt is generated. 1h = Interrupt is generated.

21.9.18 QCLR Register (Offset = 34h) [reset = 0h]

QCLR is shown in Figure 21-38 and described in Table 21-22.

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Figure 21-38. QCLR Register

15	14	13	12	11	10	9	8
RESERVED				UTO	IEL	SEL	PCM
R-0h				R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PCR	PCO	PCU	WTO	QDC	PHE	PCE	INT
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 21-22. QCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Always write as 0.
11	UTO	R/W	0h	Clear unit time out interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
10	IEL	R/W	0h	Clear index event latch interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
9	SEL	R/W	0h	Clear strobe event latch interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
8	PCM	R/W	0h	Clear eQEP compare match event interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
7	PCR	R/W	0h	Clear position-compare ready interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
6	PCO	R/W	0h	Clear position counter overflow interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
5	PCU	R/W	0h	Clear position counter underflow interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
4	WTO	R/W	0h	Clear watchdog timeout interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
3	QDC	R/W	0h	Clear quadrature direction change interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
2	PHE	R/W	0h	Clear quadrature phase error interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.
1	PCE	R/W	0h	Clear position counter error interrupt flag. 0h = No effect. 1h = Clears the interrupt flag.

Table 21-22. QCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	INT	R/W	0h	Global interrupt clear flag. 0h = No effect. 1h = Clears the interrupt flag and enables further interrupts to be generated if an event flags is set to 1.

21.9.19 QFRC Register (Offset = 36h) [reset = 0h]

QFRC is shown in Figure 21-39 and described in Table 21-23.

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Figure 21-39. QFRC Register

15	14	13	12	11	10	9	8
RESERVED				UTO	IEL	SEL	PCM
R-0h				R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
PCR	PCO	PCU	WTO	QDC	PHE	PCE	RESERVED
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R-0h

Table 21-23. QFRC Register Field Descriptions

Bit	Field	Type	Reset	Description
15-12	RESERVED	R	0h	Always write as 0s
11	UTO	R/W	0h	Force unit time out interrupt. 0h = No effect. 1h = Force the interrupt.
10	IEL	R/W	0h	Force index event latch interrupt. 0h = No effect. 1h = Force the interrupt.
9	SEL	R/W	0h	Force strobe event latch interrupt. 0h = No effect. 1h = Force the interrupt.
8	PCM	R/W	0h	Force position-compare match interrupt. 0h = No effect. 1h = Force the interrupt.
7	PCR	R/W	0h	Force position-compare ready interrupt. 0h = No effect. 1h = Force the interrupt.
6	PCO	R/W	0h	Force position counter overflow interrupt. 0h = No effect. 1h = Force the interrupt.
5	PCU	R/W	0h	Force position counter underflow interrupt. 0h = No effect. 1h = Force the interrupt.
4	WTO	R/W	0h	Force watchdog time out interrupt. 0h = No effect. 1h = Force the interrupt.
3	QDC	R/W	0h	Force quadrature direction change interrupt. 0h = No effect. 1h = Force the interrupt.
2	PHE	R/W	0h	Force quadrature phase error interrupt. 0h = No effect. 1h = Force the interrupt.
1	PCE	R/W	0h	Force position counter error interrupt. 0h = No effect. 1h = Force the interrupt.
0	RESERVED	R	0h	Always write as 0.

21.9.20 QEPSTS Register (Offset = 38h) [reset = 8Eh]

QEPSTS is shown in Figure 21-40 and described in Table 21-24.

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Figure 21-40. QEPSTS Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
UPEVNT	FIDF	QDF	QDLF	COEF	CDEF	FIMF	PCEF
R-1h	R-0h	R-0h	R-0h	R/W-1h	R/W-1h	R/W-1h	R-0h

Table 21-24. QEPSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
15-8	RESERVED	R	0h	Always write as 0.
7	UPEVNT	R	1h	Unit position event flag. 0h = No unit position event is detected. 1h = Unit position event is detected. Write 1 to clear.
6	FIDF	R	0h	Direction on the first index marker. Status of the direction is latched on the first index event marker. 0h = Counter-clockwise rotation (or reverse movement) on the first index event. 1h = Clockwise rotation (or forward movement) on the first index event.
5	QDF	R	0h	Quadrature direction flag. 0h = Counter-clockwise rotation (or reverse movement). 1h = Clockwise rotation (or forward movement).
4	QDLF	R	0h	eQEP direction latch flag. Status of direction is latched on every index event marker. 0h = Counter-clockwise rotation (or reverse movement) on index event marker. 1h = Clockwise rotation (or forward movement) on index event marker.
3	COEF	R/W	1h	Capture overflow error flag. 0h = Sticky bit, cleared by writing 1. 1h = Overflow occurred in eQEP Capture timer (QCTMR).
2	CDEF	R/W	1h	Capture direction error flag. 0h = Sticky bit, cleared by writing 1. 1h = Direction change occurred between the capture position event.
1	FIMF	R/W	1h	First index marker flag. 0h = Sticky bit, cleared by writing 1. 1h = Set by first occurrence of index pulse.
0	PCEF	R	0h	Position counter error flag. This bit is not sticky and it is updated for every index event. 0h = No error occurred during the last index transition. 1h = Position counter error.

21.9.21 QCTMR Register (Offset = 3Ah) [reset = 0h]

QCTMR is shown in Figure 21-41 and described in Table 21-25.

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Figure 21-41. QCTMR Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QCTMR															
R/W-0h															

Table 21-25. QCTMR Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QCTMR	R/W	0h	This register provides time base for edge capture unit.

21.9.22 QCPRD Register (Offset = 3Ch) [reset = 0h]

QCPRD is shown in Figure 21-42 and described in Table 21-26.

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Figure 21-42. QCPRD Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QCPRD															
R/W-0h															

Table 21-26. QCPRD Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QCPRD	R/W	0h	This register holds the period count value between the last successive eQEP position events.

21.9.23 QCTMRLAT Register (Offset = 3Eh) [reset = 0h]

QCTMRLAT is shown in Figure 21-43 and described in Table 21-27.

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Figure 21-43. QCTMRLAT Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QCTMRLAT															
R-0h															

Table 21-27. QCTMRLAT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QCTMRLAT	R	0h	The eQEP capture timer value is latched into this register on two events viz., unit timeout event, reading the eQEP position counter.

21.9.24 QCPRDLAT Register (Offset = 40h) [reset = 0h]

QCPRDLAT is shown in Figure 21-44 and described in Table 21-28.

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Figure 21-44. QCPRDLAT Register

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QCPRDLAT															
R/W-0h															

Table 21-28. QCPRDLAT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	QCPRDLAT	R/W	0h	eQEP capture period value is latched into this register on two events, unit timeout event, reading the eQEP position counter.